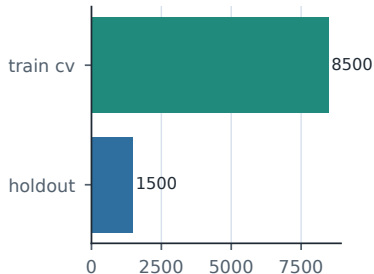


Scenario balance and leakage diagnostics

Split



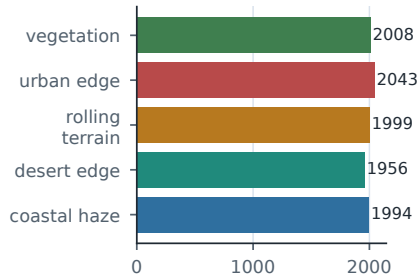
Leakage checks

canary test: detected and blocked

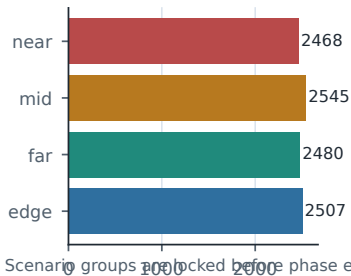
stratum baseline AP: 0.005

metadata baseline AP: 0.004

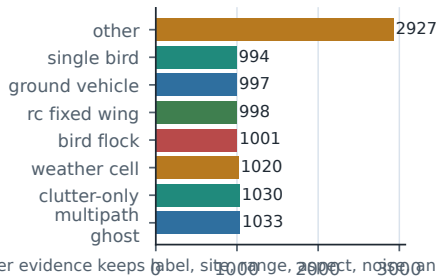
label shuffle AP: 0.006



Range



Hard-negative



Marginal imbalance

aspect: 0.000

hard_negative_role: 0.032

noise: 0.000

range: 0.000

site: 0.000

split: 0.390

Scenario groups are blocked before phase expansion. The paper evidence keeps label, site, range, aspect, noise and hard-negative family counts together.